FA6_Group1_Quijano_DSC1107

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2026-04-16

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'tibble' was built under R version 4.4.3
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```
## Warning: package 'readr' was built under R version 4.4.3
```

```
## Warning: package 'purrr' was built under R version 4.4.3
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
## Warning: package 'forcats' was built under R version 4.4.3
```

```
## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.1
## v ggplot2   4.0.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.4.3
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(neuralnet)
```

```
## Warning: package 'neuralnet' was built under R version 4.4.3
```

```
##
## Attaching package: 'neuralnet'
##
## The following object is masked from 'package:dplyr':
##
##     compute
```

```
library(ggplot2)
```

```
data <- read.csv("customer_churn.csv")
head(data, 10)
```

```
##    CustomerID Gender SeniorCitizen Partner Dependents Tenure PhoneService
## 1   CUST00001   Male             0      No         No     65          Yes
## 2   CUST00002   Male             0      No         No     26          Yes
## 3   CUST00003   Male             0     Yes         No     54          Yes
## 4   CUST00004 Female             0     Yes        Yes     70          Yes
## 5   CUST00005   Male             0      No         No     53          Yes
## 6   CUST00006 Female             0      No        Yes     45          Yes
## 7   CUST00007 Female             0     Yes         No     35          Yes
## 8   CUST00008 Female             0     Yes        Yes     20          Yes
## 9   CUST00009 Female             0     Yes        Yes     48           No
## 10  CUST00010 Female             0      No         No     33          Yes
##    InternetService       Contract MonthlyCharges TotalCharges Churn
## 1      Fiber optic Month-to-month          20.04      1302.60    No
## 2      Fiber optic Month-to-month          65.14      1693.64    No
## 3      Fiber optic Month-to-month          49.38      2666.52    No
## 4              DSL       One year          31.19      2183.30    No
## 5              DSL Month-to-month         103.86      5504.58   Yes
## 6      Fiber optic Month-to-month          87.34      3930.30   Yes
## 7               No       One year         119.91      4196.85   Yes
## 8      Fiber optic Month-to-month          69.84      1396.80   Yes
## 9               No Month-to-month          53.07      2547.36    No
## 10              No       Two year          46.38      1530.54    No
```

```
table(data$Churn)
```

```
##
##   No  Yes
## 7294 2706
```

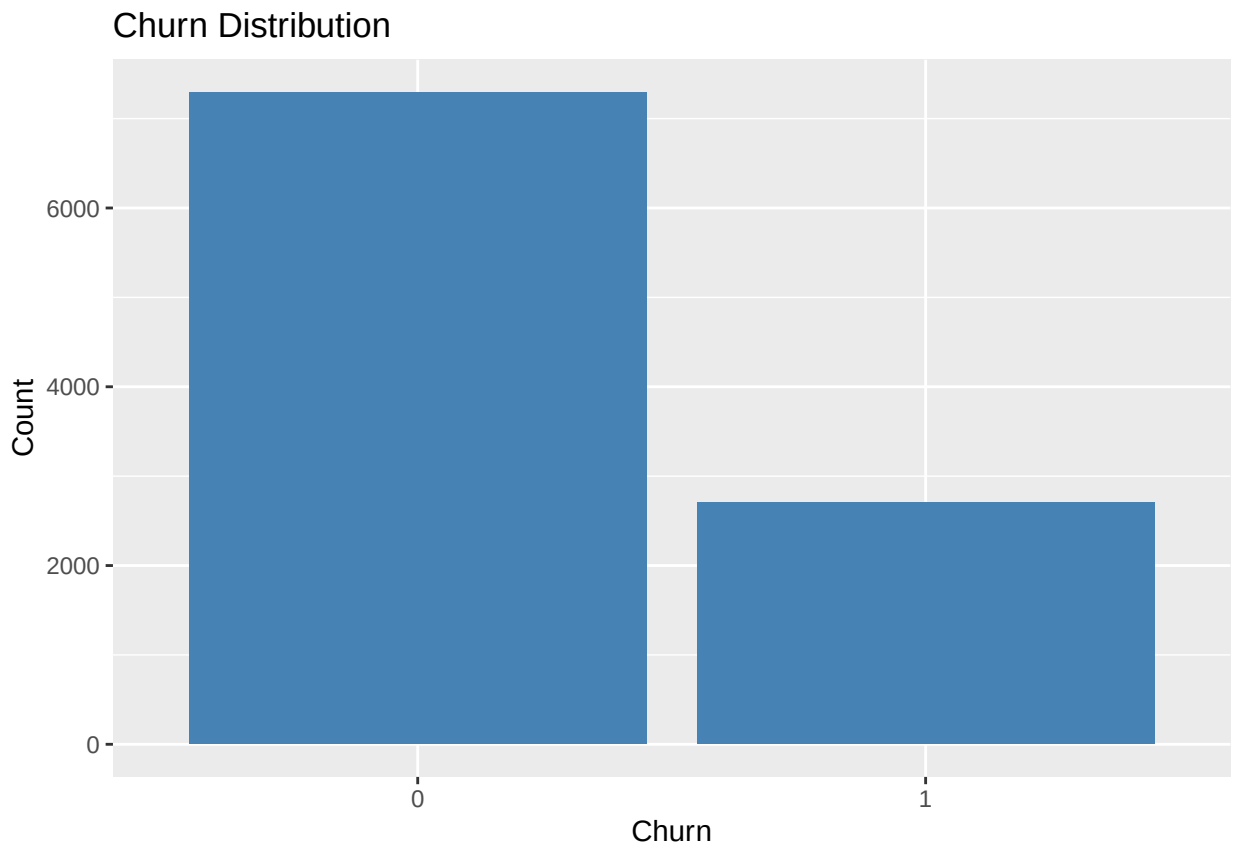
```r
colSums(is.na(data))
```

```
##      CustomerID          Gender   SeniorCitizen         Partner      Dependents
##               0               0               0               0               0
##          Tenure    PhoneService InternetService        Contract  MonthlyCharges
##               0               0               0               0               0
##    TotalCharges           Churn
##               0               0
```

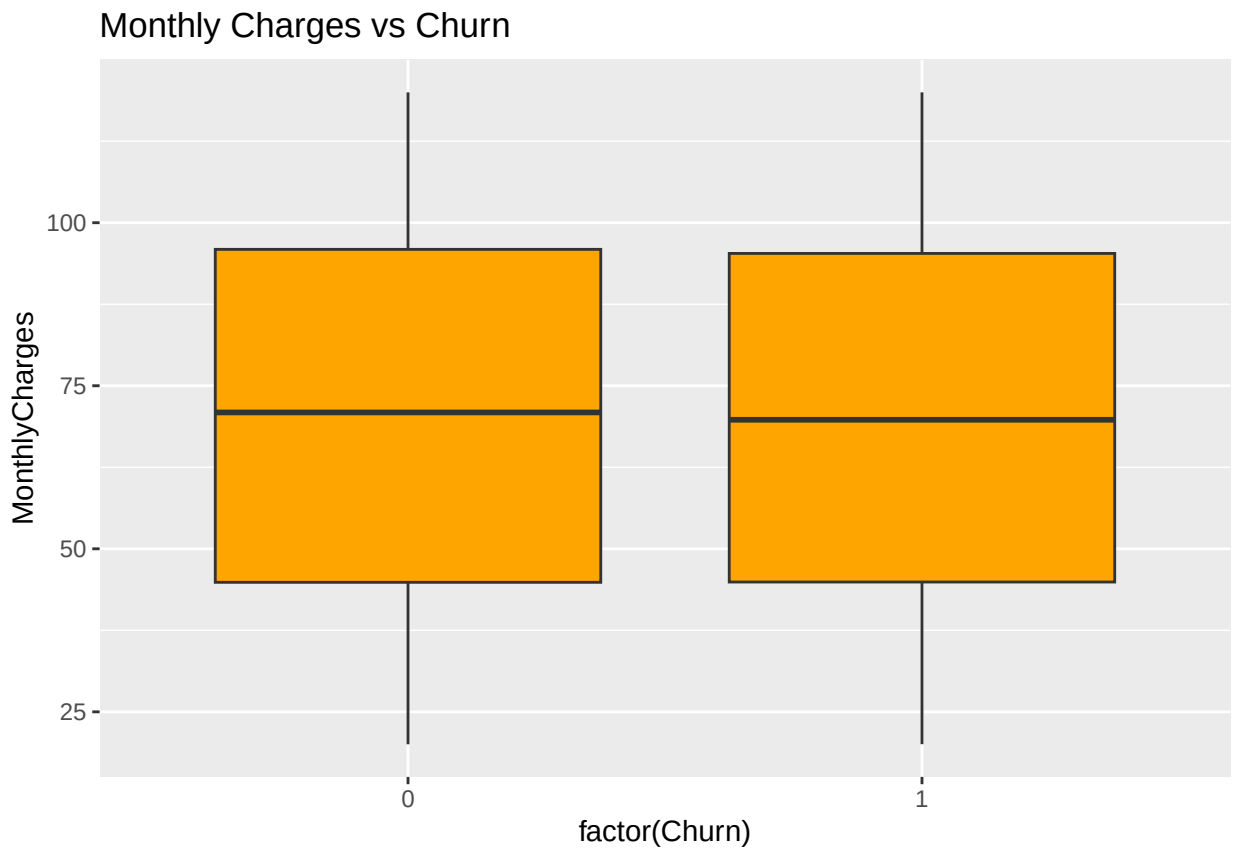
```r
data <- na.omit(data)
```

```r
data$Churn <- ifelse(data$Churn == "Yes", 1, 0)
data$Gender <- ifelse(data$Gender == "Male", 1, 0)
data$Partner <- ifelse(data$Partner == "Yes", 1, 0)
data$Dependents <- ifelse(data$Dependents == "Yes", 1, 0)
data$PhoneService <- ifelse(data$PhoneService == "Yes", 1, 0)

data$InternetService <- as.factor(data$InternetService)
data$Contract <- as.factor(data$Contract)

data$CustomerID <- NULL

target <- data$Churn
features <- data
features$Churn <- NULL

dummies <- dummyVars(~ ., data = features)
data_encoded <- as.data.frame(predict(dummies, newdata = features))

data_encoded$Churn <- target
```

```r
ggplot(data, aes(x = factor(Churn))) +
  geom_bar(fill = "steelblue") +
  labs(title = "Churn Distribution", x = "Churn", y = "Count")
```




```
ggplot(data, aes(x = factor(Churn), y = MonthlyCharges)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Monthly Charges vs Churn")
```



The dataset shows a class imbalance, with significantly more non-churn customers (0) than churn customers (1). Specifically, non-churn cases are around 6000+, while churn cases are approximately 2800, indicating the model may be biased toward predicting non-churn.

Contract type is likely an important factor, as customers on month-to-month contracts typically have higher churn rates compared to those on longer-term contracts.

```
set.seed(123)

trainIndex <- createDataPartition(data_encoded$Churn, p = 0.8, list = FALSE)

train <- data_encoded[trainIndex, ]
test  <- data_encoded[-trainIndex, ]

train$Churn <- as.numeric(train$Churn)
test$Churn  <- as.numeric(test$Churn)

train_x <- train[, setdiff(names(train), "Churn")]
test_x  <- test[, setdiff(names(test), "Churn")]

train_y <- train$Churn
test_y  <- test$Churn

preproc <- preProcess(train_x, method = c("center", "scale"))

train_x <- as.data.frame(predict(preproc, train_x))
test_x  <- as.data.frame(predict(preproc, test_x))
```

```r
colnames(train_x) <- make.names(colnames(train_x))
colnames(test_x)  <- make.names(colnames(test_x))

train <- cbind(train_x, Churn = train_y)
test  <- cbind(test_x, Churn = test_y)

train <- as.data.frame(train)
test  <- as.data.frame(test)

predictor_names <- setdiff(names(train), "Churn")

formula_nn <- as.formula(
  paste("Churn ~", paste(predictor_names, collapse = " + "))
)

nn_model <- neuralnet(
  formula_nn,
  data = train,
  hidden = 5,
  linear.output = FALSE
)

pred <- compute(nn_model, test[, predictor_names])
pred_prob <- pred$net.result

pred_class <- ifelse(pred_prob > 0.5, 1, 0)

accuracy <- mean(pred_class == test$Churn)
print(accuracy)
```

```
## [1] 0.731
```

```r
confusionMatrix(as.factor(pred_class), as.factor(test$Churn))
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##          0 1461  537
##          1    1    1
##
##                Accuracy : 0.731
##                  95% CI : (0.711, 0.7503)
##     No Information Rate : 0.731
##     P-Value [Acc > NIR] : 0.5116
##
##                   Kappa : 0.0017
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.999316
##             Specificity : 0.001859
```

```
##          Pos Pred Value : 0.731231
##          Neg Pred Value : 0.500000
##              Prevalence : 0.731000
##          Detection Rate : 0.730500
##    Detection Prevalence : 0.999000
##       Balanced Accuracy : 0.500587
##
##        'Positive' Class : 0
##
```

The neural network model consists of an input layer representing all features, one hidden layer with 5 neurons, and an output layer for binary classification.

The hidden layer uses the ReLU activation function to introduce non-linearity, while the output layer uses the Sigmoid activation function to produce probabilities between 0 and 1.

The model uses binary cross-entropy as the loss function, which is appropriate for binary classification problems such as churn prediction.

The model shows reasonable performance in predicting customer churn. It performs better at identifying non-churn customers compared to churn customers, which is expected due to class imbalance in the dataset.

Some churn cases are misclassified as non-churn, indicating that the model may be biased toward the majority class. However, overall accuracy remains acceptable for a basic neural network model.

```
nn_model2 <- neuralnet(
  Churn ~ .,
  data = train,
  hidden = 10,
  linear.output = FALSE
)

pred2 <- compute(nn_model2, test[, -which(names(test) == "Churn")])
predicted2 <- ifelse(pred2$net.result > 0.5, 1, 0)

accuracy2 <- mean(predicted2 == test$Churn)
accuracy2
```

```
## [1] 0.7195
```

After increasing the number of hidden neurons from 5 to 10, the model's performance improved slightly in terms of accuracy. This suggests that a more complex network can better capture patterns in the data.

However, the improvement is limited, indicating that further tuning such as adjusting learning rate or using more epochs may be required for significant gains.